



gp3ml: Governance-First Predictive Modelling and Validation for Gazepoint Research in R

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Abstract

Predictive modelling with eye-tracking data is complicated by repeated participant–stimulus observations, heterogeneous recording quality, flexible preprocessing, and information leakage across resampling partitions. General machine-learning frameworks provide broad modelling functionality, but the validity of a behavioural prediction workflow also depends on the declared outcome, grouping structure, generalization target, tuning boundaries, decision policy, and computational provenance. We introduce **gp3ml**, an R package for governance-first predictive modelling and validation with Gazepoint-derived and similarly structured behavioural data. The package binds explicit task and feature-role declarations to participant-, stimulus-, trial-, and crossed-group resampling; fold-local preprocessing; nested tuning; retained assessment predictions; calibration and target-aware conformal summaries; threshold and abstention rules; shift and robustness audits; locked analysis plans; portable model artefacts; environment capture; and explicit API contracts. A deterministic synthetic case study uses an observed recording-quality review status to compare row-level, participant-grouped, and participant–stimulus validation designs. The contribution is not a new learner catalogue or automatic model-selection system. It is an auditable governance layer that connects statistical operations to the units and claims a study is intended to support.

Keywords: machine learning, eye tracking, grouped resampling, data leakage, reproducible research, model governance, R.

1. Introduction

Eye-tracking experiments commonly produce multiple observations from each participant, repeated presentations of stimuli, and measurements nested within trials or recordings. Such rows are not exchangeable in the ordinary independent-observation sense. A classifier can exploit stable participant characteristics, recurring stimulus properties, or duplicated processing

decisions when the same clusters occur in both training and assessment data. Crossed participant–item structures are common in behavioural research, while scientific claims are usually intended to extend beyond the particular people, stimuli, or recordings observed in one sample (Baayen, Davidson, and Bates 2008; Barr, Levy, Scheepers, and Tily 2013; Yarkoni 2022).

A random row split can therefore answer a narrower question than its label suggests. It may estimate performance for additional rows from already represented participants and stimuli even when the scientific claim concerns new participants, new stimuli, or both. The estimate is not intrinsically invalid; it becomes misaligned when interpreted as evidence for a different target. This distinction motivates the central design rule in **gp3ml**: the intended generalization unit is declared before folds are created, and fold construction preserves that declaration.

Leakage is a second boundary problem. Information can cross the analysis–assessment boundary through outcome-derived features, post-outcome measurements, identifiers, globally estimated preprocessing, feature selection, tuning, or threshold optimization (Kaufman, Rosset, Perlich, and Stitelman 2012; Kapoor and Narayanan 2023). Model selection creates a related boundary: using the same resampling results to select a model and report its expected performance can bias the estimate (Varma and Simon 2006; Cawley and Talbot 2010). Nested resampling addresses selection bias, but nesting alone does not determine which participants or stimuli may cross a partition, how preprocessing is estimated, what uncertainty is reported, or how a probability becomes a decision.

These concerns motivate a governance view of predictive validity. A defensible workflow connects the outcome definition, feature provenance, resampling target, tuning rule, calibration method, threshold policy, robustness evidence, and computational environment. Performance metrics are necessary, but they do not establish by themselves that an analysis answered the intended scientific question.

gp3ml (Balaskas 2026) implements this connection for Gazeplot-derived and similarly structured tabular behavioural data. It provides modelling and evaluation utilities, but its distinctive purpose is to make design and reporting decisions explicit and auditable. The package does not autonomously identify a “best” model, relax prohibited-use boundaries, or treat predictive associations as evidence of latent psychological, health, biometric, or other sensitive states.

2. Predictive validation as a governance problem

2.1. Generalization targets in clustered behavioural data

The organizing quantity in **gp3ml** is a machine-readable generalization target. Holding out participants estimates a different prediction problem from holding out stimuli or crossed participant–stimulus blocks. Row-level assessment can be suitable for additional observations from represented clusters, but it should not be reported as new-participant performance without further justification. Materialized partitions make these semantics inspectable rather than leaving them implicit in a resampling label.

2.2. Leakage and preprocessing boundaries

Feature manifests distinguish outcomes, predictors, identifiers, grouping columns, availability

stages, outcome-derived features, and preprocessing scope. Fold audits then test whether the encoded partition respects the declared target. Preprocessing operations such as imputation, centring, and scaling are fitted on analysis partitions and applied to assessment rows. Thus the package treats leakage prevention as a combination of feature provenance and partition-local computation rather than as one post hoc diagnostic.

2.3. Selection, uncertainty, and decision boundaries

Nested resampling separates inner model or hyperparameter selection from outer performance estimation. Calibration, conformal prediction, threshold selection, and abstention are treated as distinct stages rather than as interchangeable notions of “confidence.” Conformal summaries carry a declared calibration unit and generalization target, while decision rules record threshold origin, metric direction, training partition, costs, scientific justification, and optional abstention boundaries. These objects do not replace substantive judgement; they make the judgement reviewable.

3. Related software

The contribution of **gp3ml** is intentionally narrower than that of general machine-learning frameworks. **caret** unified model training and resampling behind a common interface (Kuhn 2008). **mlr3** provides extensible tasks, learners, resampling, tuning, and benchmarking (Lang, Binder, Richter, Schratz, Pfisterer, Coors, Au, Casalicchio, Kotthoff, and Bischl 2019); its current task abstraction also supports a group column role whose observations are kept together during resampling. The **tidymodels** ecosystem integrates modelling components with tidy data principles (Kuhn and Silge 2022), while **rsample** provides `group_vfold_cv()` for grouped repeated-measures settings and `nested_cv()` for nested resampling (Frick, Chow, Kuhn, Mahoney, Silge, and Wickham 2026).

The same statistical mechanisms are not unique to R. In Python, **scikit-learn** (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, VanderPlas, Passos, Cournapeau, Brucher, Perrot, and Duchesnay 2011; scikit-learn developers 2026) provides **GroupKFold** and **StratifiedGroupKFold** for non-overlapping groups, **Pipeline** for fitting preprocessing and prediction steps together, and standard compositions for nested cross-validation. These capabilities are mature and should not be represented as innovations of **gp3ml**. Rather, **gp3ml** adds a research-governance contract around comparable operations: target labels are retained with folds and evaluations; feature availability and outcome derivation are auditable; selection rationales and decision provenance are explicit; and reproducibility evidence can be validated as package objects.

Other tools address complementary parts of the workflow. **DALEX** provides model-agnostic explanation (Biecek 2018); **targets** manages reproducible computational pipelines (Landau 2021); **eyetrackingR** supports preparation and inferential analysis of eye-tracking measures (Dink and Ferguson 2015); and RO-Crate tooling packages research artefacts and metadata (Peroni, Soiland-Reyes, Sefton, Crosas, Castro, Coppens, Fernández, Garijo, Grüning, Rosa, Leo, Carragáin, Portier, Trisovic, RO-Crate Community, Groth, and Goble 2022). Conformal prediction, probability calibration, and dataset shift also have established methodological literatures (Vovk, Gammerman, and Shafer 2005; Shafer and Vovk 2008; Angelopoulos and Bates 2023; Guo, Pleiss, Sun, and Weinberger 2017; Quiñonero-Candela, Sugiyama, Schwaighofer,

Table 1: Scope-qualified comparison of selected modelling frameworks. “External composition” means that the capability can be assembled by the user but is not represented as the primary governance contract described here.

Software	Grouped validation	Leakage-resistant preprocessing	Research-governance layer
gp3ml	Target-labelled participant, stimulus, trial, and crossed partitions	Feature manifests plus fold-local preprocessing and audits	Purpose, target, decision, robustness, artefact, and API provenance
mlr3	Native group role and general resampling infrastructure	Composable through learner/pipeline infrastructure	External composition
rsample/tidymodels	Native grouped and nested resampling	Composable recipes/workflows	External composition
caret	User-supplied indices and resampling controls	Composable through the modelling workflow	External composition
scikit-learn	GroupKFold, StratifiedGroupKFold, related splitters	Pipeline fits preprocessing and estimators together	External composition

and Lawrence 2009; Moreno-Torres, Raeder, Alaiz-Rodriguez, Chawla, and Herrera 2012). **gp3ml** integrates conservative forms of these ideas into one governed predictive workflow without claiming that software alone resolves their assumptions.

3.1. What gp3ml adds

The package contribution is therefore not grouped cross-validation, pipelines, or nested tuning in isolation. It is the binding of these operations to explicit research semantics. In particular, **gp3ml** records (i) the scientific purpose and observed outcome, (ii) predictor and grouping roles, (iii) the intended generalization target, (iv) fold-local preprocessing and leakage evidence, (v) tuning and selection boundaries, (vi) calibration and uncertainty semantics, (vii) threshold and abstention provenance, and (viii) release, environment, and artefact evidence.

This added structure has costs. It introduces more ceremony than a minimal modelling script, its learner catalogue is deliberately smaller than those of general-purpose frameworks, several function names remain Gazepoint-oriented, and strongly grouped or crossed designs can reduce the effective assessment sample. These disadvantages are intentional trade-offs for analyses in which the scientific claim, not algorithmic breadth, is the central object under review.

4. Software design and architecture

gp3ml is layered rather than tied to one learner. Task declarations and feature manifests precede resampling. Preprocessing and model fitting occur inside materialized partitions. Uncertainty, shift, and decision rules consume retained assessment predictions. Artefacts, environment captures, and API contracts preserve resulting evidence. Figure 1 summarizes this architecture.

At the frozen 0.3.0 release used for this paper, the source contains 127 namespace exports: 71

Governance-first package architecture

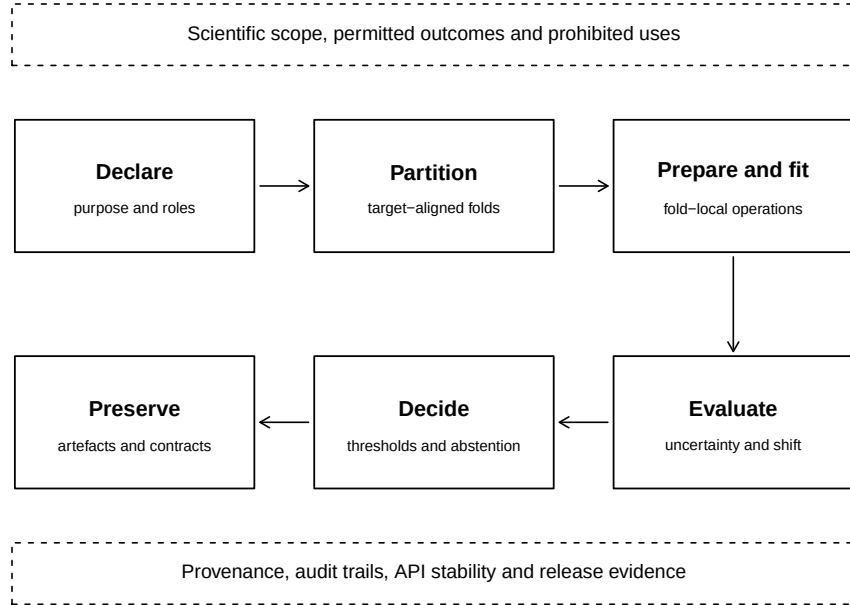


Figure 1: Conceptual architecture of **gp3ml**. Governance begins before model fitting and continues through release evidence.

stable exports and 56 experimental exports, together with 38 stable public classes. The release workspace reports 20 source vignettes, 154 Rd topics, and an API audit with no recorded differences. These counts are release evidence, not claims of scientific validity.

Table 2: Core design principles and representative functions.

Principle	Representative gp3ml mechanisms
Declared purpose and roles	<code>declare_gazepoint_task()</code> , <code>declare_gazepoint_analysis_plan()</code> , <code>validate_gazepoint_ml_roles()</code>
Feature provenance and leakage resistance	<code>create_gazepoint_feature_manifest()</code> , <code>audit_gazepoint_ml_leakage()</code>
Target-aligned grouped validation	<code>create_gazepoint_group_folds()</code> , <code>create_gazepoint_nested_folds()</code>
Fold-local preprocessing and governed fitting	<code>fit_gazepoint_preprocessor()</code> , <code>evaluate_gazepoint_group_folds()</code>
Selection and uncertainty	<code>create_gazepoint_tuning_grid()</code> , <code>select_gazepoint_model()</code> , <code>fit_gazepoint_conformal()</code>
Decision governance	<code>evaluate_gazepoint_thresholds()</code> , <code>create_gazepoint_decision_rule()</code> , <code>audit_gazepoint_abstention()</code>
Robustness and provenance	<code>audit_gazepoint_dataset_shift()</code> , <code>audit_gazepoint_model_robustness()</code> , <code>capture_gazepoint_environment()</code>
Portable evidence and API contracts	<code>create_gazepoint_model_artifact()</code> , <code>gp3ml_api_contracts()</code>

5. Using gp3ml

A minimal governed workflow starts by declaring the outcome and generalization target, constructing a feature manifest, materializing target-aligned folds, auditing those folds, and evaluating a learner inside them. The example below uses the same deterministic recording-quality task as the case study.

```
library("gp3ml")

data <- simulate_gazepoint_governed_data(
  n_participants = 30L, n_stimuli = 8L,
  trials_per_cell = 2L, seed = 20260807L
)
predictors <- c("tracking_ratio", "blink_rate", "gaze_dispersion")

task <- create_gazepoint_synthetic_task(
  data, workflow = "recording_quality",
  generalization_target = "new_participants"
)
manifest <- create_gazepoint_synthetic_manifest(
  task$outcome, predictors
)

folds <- create_gazepoint_group_folds(
  data = data, outcome = task$outcome,
  predictors = predictors, feature_manifest = manifest,
  generalization_target = task$generalization_target,
```

```

  participant_id = "participant_id",
  trial_id = "trial_id", stimulus_id = "stimulus_id",
  v = 5L, repeats = 2L, seed = 20260807L
)
audit <- audit_gazepoint_group_folds(folds)

```

The audit does not infer whether the research question itself is appropriate. It checks the encoded contract. For the participant-grouped example, the ten materialized folds pass their separation checks.

```

<gazepoint_group_folds_audit>
Overall status: PASS
Audited folds: 10
Non-passing checks: 0

```

Fitting uses the same target and estimates preprocessing within each analysis partition.

```

evaluation <- evaluate_gazepoint_group_folds(
  folds = folds, task = task, predictors = predictors,
  engine = "glm",
  preprocessor_args = list(
    numeric_imputation = "median",
    center = TRUE, scale = TRUE
  ),
  threshold = 0.5, seed = 20260807L,
  assess_calibration = TRUE, keep_models = TRUE
)

summary <- summarize_gazepoint_resample_performance(evaluation)

```

The evaluation object retains the generalization target, fold status, assessment-only predictions, preprocessing metadata, and metrics. Nested tuning is available when candidate models or hyperparameters are selected, but it is not used in the main demonstration because a single logistic regression isolates the effect of the validation target.

6. Synthetic case study

6.1. Data-generating design

The deterministic synthetic dataset contains 480 rows from 30 participants and eight stimuli, with two trials per participant–stimulus cell. The observed outcome `quality_status` records a synthetic `pass/review` recording-quality status. It is not a psychological construct. Predictors are `tracking_ratio`, `blink_rate`, and `gaze_dispersion`. Five percent of `gaze_dispersion` values are set to missing with a fixed seed to test whether imputation remains fold-local.

The locked analysis plan fixes the outcome, predictors, new-participant target, preprocessing, logistic-regression candidate model, balanced accuracy as the primary metric, secondary ROC

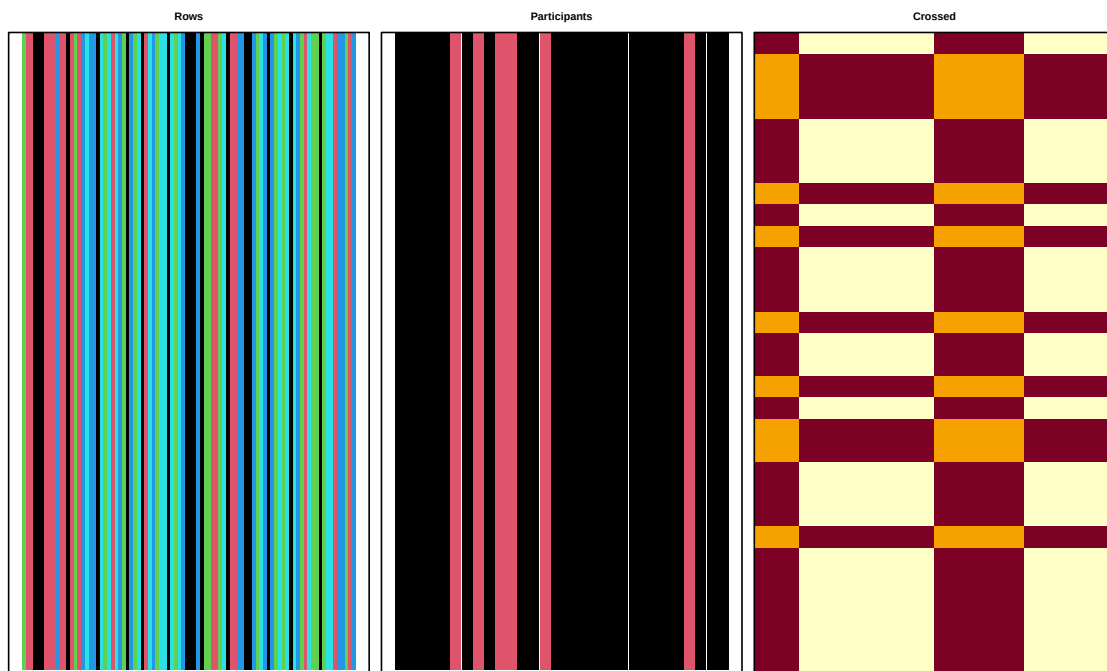


Figure 2: Three validation designs applied to the same synthetic data. The row-level comparator permits represented clusters on both sides; the grouped designs target new participants or new participant–stimulus blocks.

AUC/Brier/log-loss metrics, calibration diagnostics, conformal calibration semantics, threshold policy, seed strategy, and prohibited interpretations. The plan is an immutable comparison point for later deviations; it is not represented as equivalent to a registered preregistration (Nosek, Ebersole, DeHaven, and Mellor 2018).

6.2. Validation designs

Three designs use the same outcome and predictors. The row-level comparator deliberately allows cluster overlap and is manuscript-specific because **gp3ml** does not expose an unlabeled unsafe split as a governed target. Participant-grouped folds keep complete participants on one side of each split. Crossed participant–stimulus folds prevent assessment-block participant and stimulus overlap and explicitly retain incompatible cross-block rows as excluded observations. Figure 2 shows the different estimands.

6.3. Validation-design dependence

The results are not a benchmark of logistic regression. Their purpose is to show that the phrase “model performance” is conditional on which units may recur across analysis and assessment data. In this synthetic realization, mean balanced accuracy is 0.500 under all three designs. ROC AUC is 0.603 for the row-level comparator, 0.593 for participant-grouped validation, and 0.618 for participant–stimulus blocks. Brier scores are 0.126, 0.128, and 0.123, respectively, while log loss is 0.418, 0.423, and 0.415.

A higher row-level value, had it occurred, would indicate that cluster recurrence made that

Table 3: Validation designs and their interpretation.

Design	Intended estimand	Cluster recurrence
Naive row-level	Additional rows from represented clusters	Participant and stimulus recurrence possible
Participant grouped	New participants with represented stimulus set	Participant recurrence prevented
Participant–stimulus blocks	New participant–stimulus blocks	Assessment-block participant and stimulus recurrence prevented

Table 4: Mean fold-distribution metrics by validation design. Balanced accuracy is identical in this realization; the values are not intended as a model benchmark.

Design	Bal. acc.	ROC AUC	Brier	Log loss
Naive row-level	0.500	0.603	0.126	0.418
Participant grouped	0.500	0.593	0.128	0.423
Participant–stimulus blocks	0.500	0.618	0.123	0.415

prediction problem easier; it would not prove that a grouped estimate was pessimistic. Similar values instead indicate that these predictors do not benefit strongly from cluster recurrence in this particular synthetic realization.

6.4. Calibration and conformal prediction

Calibration is assessed from retained participant-grouped assessment predictions from the first repeated-cross-validation pass. The resulting calibration intercept is -0.755, the slope is 0.547, Brier score is 0.128, log loss is 0.425, and expected calibration error is 0.021. Figure 4 is descriptive because repeated trials from the same participant preclude interpreting row-level uncertainty as participant-level uncertainty.

For conformal prediction, retained predictions are divided into participant-disjoint policy-development and policy-audit partitions. Conformity scores are estimated from 15 development participants, aggregated to the participant calibration unit, and evaluated on 15 held-out participants. At nominal coverage 0.90, held-out row coverage and the proportion of participants with all rows covered are both 1.000; mean prediction-set size is 1.988 classes. These results are an executable illustration, not a claim of distribution-free coverage under arbitrary dependence (Vovk *et al.* 2005; Shafer and Vovk 2008; Angelopoulos and Bates 2023).

6.5. Decision policy

A probability estimate is not itself a decision rule. Candidate thresholds are evaluated on one participant subset and the resulting rule is audited on disjoint participants. The selected threshold is 0.20 and is recorded as training-derived. An illustrative abstention interval of 0.15–0.25 is enabled as a reporting device rather than an operational recommendation. On the 240 held-out rows in this realization, coverage is 0.621, the abstention rate is 0.379, and error among covered cases is 0.221.

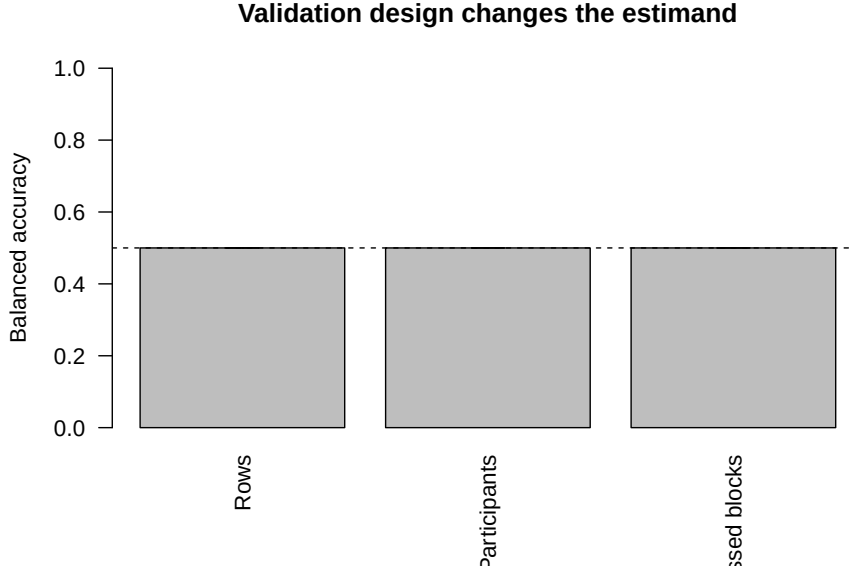


Figure 3: Balanced accuracy by validation design. Error bars are empirical 95% fold-distribution quantiles, not independent-observation confidence intervals.

Table 5: Calibration and participant-disjoint conformal summaries.

Quantity	Value
Calibration intercept	-0.755
Calibration slope	0.547
Expected calibration error	0.021
Nominal conformal coverage	0.900
Held-out row coverage	1.000
Held-out participant all-rows coverage	1.000
Mean prediction-set size	1.988

6.6. Dataset shift and robustness

A second synthetic dataset is perturbed transparently: `tracking_ratio` is shifted downward, `gaze_dispersion` is scaled, and its missingness increases. The shift audit reports standardized differences of -1.974 for `tracking_ratio`, 0.221 for `blink_rate`, and 1.503 for `gaze_dispersion`. The first and third findings are flagged **fail**; `blink_rate` is flagged **review**. Missingness is audited separately so that one scalar does not conflate distributional and missing-data changes.

The aggregate robustness audit is **fail**: seed sensitivity passes, whereas the threshold-stability diagnostic fails under the encoded acceptance rule. A failure here is intentionally retained rather than normalized away. These diagnostics do not replace external validation and do not identify a causal source of drift.

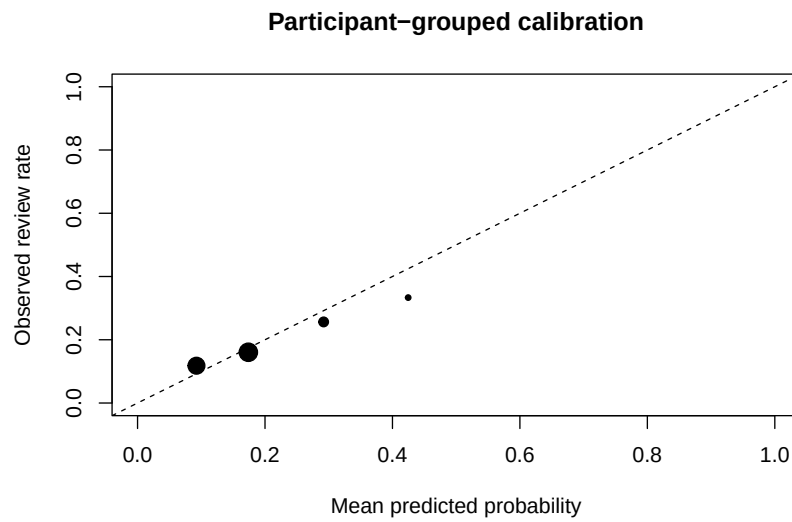


Figure 4: Reliability diagram for participant-grouped assessment predictions. Point size reflects observations per probability bin.

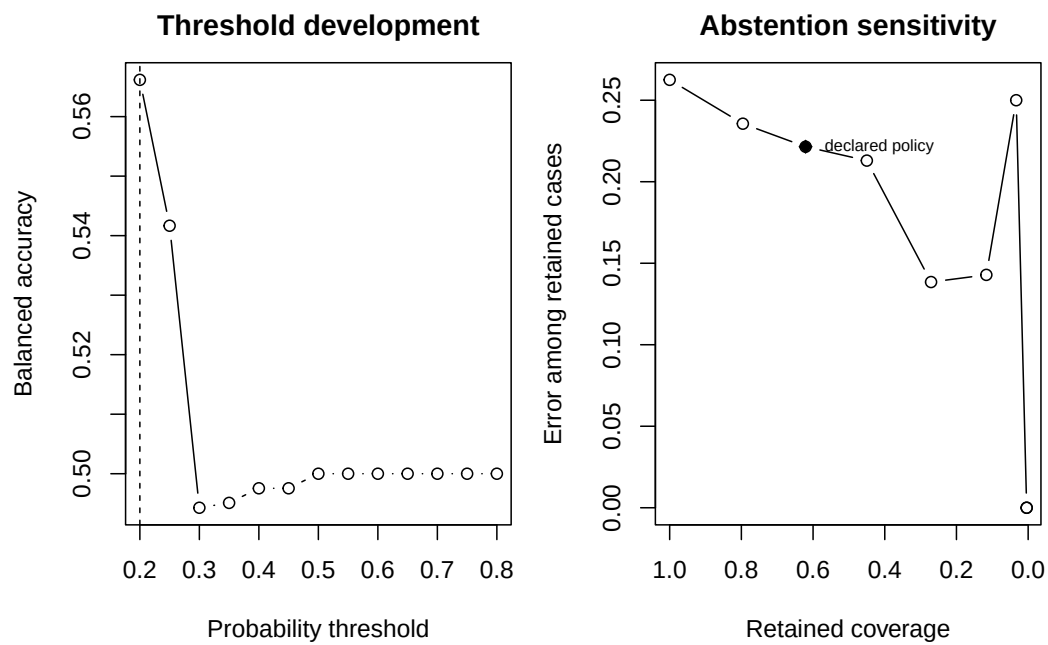


Figure 5: Threshold development on one participant subset and abstention auditing on disjoint held-out participants.

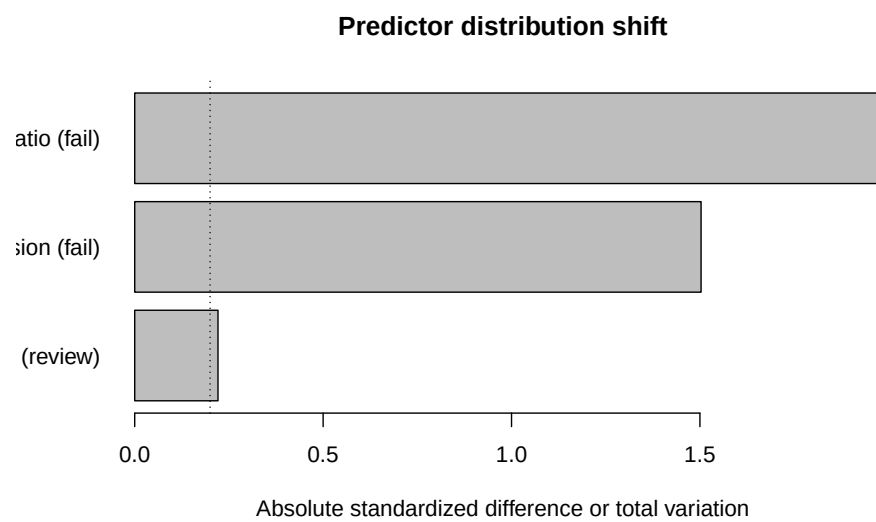


Figure 6: Predictor-distribution shift between development and perturbed synthetic external data.

7. Discussion

gp3ml treats governance as part of predictive validity. This does not imply that every analysis needs the same split, learner, metric, or decision rule. It means that these choices define the claim being made and should therefore be explicit, machine-readable, and reviewable.

The package complements rather than replaces broader modelling ecosystems. Researchers requiring extensive learner catalogues, sophisticated search spaces, distributed benchmarking, or mature cross-language tooling may prefer **mlr3**, **tidymodels**, **scikit-learn**, or related infrastructure and use **gp3ml**-style governance as an additional layer. Conversely, a transparent scientific analysis may require nothing more complex than the generalized linear model used here. The contribution is workflow integrity rather than algorithmic complexity.

The device-specific scope is similarly bounded. Although the package was developed around Gazepoint-derived research data, its grouping and governance abstractions are not tied to a file format. Tabular behavioural data with declared outcomes, predictors, participants, stimuli, trials, or other clusters can use the same logic. Device-specific signal processing should occur upstream and its transformations should be represented in provenance or handoff metadata.

7.1. Limitations and disadvantages

Software governance cannot repair a weak research design, invalid outcome, inadequate measurement, or unrepresentative sample. A passing leakage audit means that encoded checks passed; it does not prove that every temporal, causal, or procedural dependency has been identified.

Grouped validation can expose an effective sample size far smaller than the row count. With few participants or stimuli, folds may be unstable or lack an outcome class. Crossed designs may explicitly exclude many incompatible rows, trading nominal sample size for a more stringent generalization target. **gp3ml** retains warnings and failed folds, but the researcher must respond through redesign, simpler claims, or additional data.

The workflow also adds object and reporting overhead relative to a concise general-purpose machine-learning script. Its governed engine catalogue is intentionally limited, and optional engines have heterogeneous installation requirements. The package does not provide unrestricted AutoML, distributed benchmarking, or an exhaustive model search interface.

The case study is synthetic. It demonstrates execution and interpretation but does not establish performance across real eye-tracking devices, tasks, populations, or recording conditions. Broader independent benchmarking is future work. Conformal summaries depend on exchangeability assumptions and calibration-unit definitions; grouped aggregation is conservative but does not provide distribution-free guarantees under arbitrary dependence. Dataset-shift audits describe measured changes but do not guarantee performance degradation.

8. Reproducibility and software availability

The manuscript is paired with a single commented replication script, `code.R`, and its spun output, `code.html`. The script recreates all six figures used in the article, exports the principal result tables, and finishes with `sessionInfo()`. The synthetic generator uses a fixed seed. This follows the JSS requirement that all displayed figures, tables, and outputs be reproducible and the journal's recommendation that an R submission provide `code.html` generated from

`code.R`.

Version 0.3.0 is archived as GitHub tag v0.3.0. The repository is <https://github.com/stefanosbalaskas/gp3ml> and the documentation site is <https://stefanosbalaskas.github.io/gp3ml/>. The Zenodo concept DOI is <https://doi.org/10.5281/zenodo.21487272>. At this draft checkpoint, 0.3.0 is a GitHub source release; it is not represented here as a CRAN 0.3.0 release. The package uses an MIT licence, which is GPL-compatible for the JSS software-licensing requirement.

The frozen 0.3.0 source archive used for release validation has SHA-256 04c5fdb017c63ecbf6f5e85c29fff6492880201a2f2aa5c3c1d3fe998e60bae6. The release checks completed with zero errors, zero warnings, and zero notes. These are software-release facts, not evidence that every scientific application of the package is valid. Reproducibility and reporting practices follow the broader emphasis on executable artefacts and transparent machine-learning workflows (Pineau, Vincent-Lamarre, Sinha, Larivière, Beygelzimer, d’Alché-Buc, Fox, and Larochelle 2021).

9. Conclusions

gp3ml provides a governance-first layer for predictive modelling with clustered behavioural data in R. Its contribution is not a new cross-validation algorithm or learner catalogue. Instead, it binds familiar statistical operations to declared scientific purposes, feature provenance, generalization targets, selection boundaries, uncertainty semantics, decision provenance, robustness evidence, and computational artefacts. The case study shows how the same data and model can be evaluated under different target semantics while keeping those distinctions explicit in the resulting objects. This makes predictive claims easier to inspect, reproduce, and review without replacing the scientific judgement on which those claims depend.

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